

Sardor Sobirov

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EDUCATION

University of California San Diego (UCSD)

La Jolla, CA

Bachelor of Science in Data Science

Expected Jun 2026

- **Relevant Coursework:** Machine Learning Algorithms, Deep Learning and Neural Networks, Probabilistic AI and Graphical Models, Data Mining and Statistical Learning, Distributed Data Systems and Scalable Analytics.

SKILLS

Languages: Python (pandas, NumPy, scikit-learn, PyTorch), SQL, JavaScript, HTML/CSS.

Data & Analytics: A/B Testing, Hypothesis Testing, Funnel Analysis, KPI Design, Metric Trees, Exploratory Data Analysis.

Data Platforms & Cloud: BigQuery, Databricks, AWS, PostgreSQL.

Visualization: Tableau, Power BI, Matplotlib, Plotly, ArcGIS.

EXPERIENCE

Cotality

Sep 2025 – Mar 2026

HDSI Capstone Researcher

San Diego, CA

- Built a React/TypeScript geospatial dashboard using MapLibre GL, D3.js, Vite, and Tailwind CSS for Cotality's data science team to compare county-level property-characteristic distributions across California
- Used BigQuery GIS, SQL, H3 spatial indexing, spatial joins, buffers, and centroid-based geometry processing to prepare property and land-cover data for county-level anomaly detection

Apptics.ai

Jun 2025 – Sep 2025

Data Engineer Intern

Walnut Creek, CA

- Built a reproducible Python ETL pipeline ingesting daily paid media performance data from the Meta Ads API into PostgreSQL for 5 DTC brands, consolidating 1K-10K/day per brand spend into a single source of truth for reporting.
- Designed an analytics-ready relational schema (campaign/ad set/ad levels) and standardized 25+ marketing KPIs, using SQL and PowerBI, enabling consistent cross-brand performance tracking toward higher-converting ad sets.
- Ran A/B tests on Meta ad creatives using Python and Tableau, then translated statistical findings into plain-language recommendations for non-technical marketing teams contributing to a 35% improvement in acquisition efficiency.

United States Department of Defense (DoD)

Sep 2024 – Dec 2024

Product Manager Intern

La Jolla, CA

- Conducted 50+ stakeholder interviews using Lean LaunchPad methodology to translate ambiguous user needs into a prioritized product roadmap, ensuring R&D investments were tied to concrete operational use cases.
- Designed a portfolio analytics dashboard in Excel and Power BI mapping technology maturity signals to DoD use cases; presented findings to cross-functional leadership to inform multi-million dollar transition decisions.

PROJECTS

Spatial Bayesian Analysis of Transit Accessibility

Interactive Dashboard

- Fit a BYM2 Bayesian spatial hierarchical model (ICAR partial pooling, Student-t likelihood, PC priors) in PyMC across 726 San Diego census tracts, 32,000 MCMC draws, 4 chains, zero divergences. Produce full posterior distributions of transit accessibility rather than point estimates; posterior exceedance analysis uncovered a systematic density confound that deterministic methods miss.
- Designed a probabilistic intervention simulation comparing Bayesian targeting, routes prioritized by posterior exceedance probability, against deterministic targeting (prioritized by point-estimate job counts) across 720 tracts, Bayesian prioritization produced 50% more crossings from desert to adequate transit access.

Wildfire Property Intelligence: Geospatial Data Quality & Anomaly Detection

Interactive Dashboard

- Built a multi-method spatial anomaly detection pipeline over 58 California counties combining Jensen-Shannon Divergence, Moran's I spatial autocorrelation, and a CatBoost classifier two-sample test (C2ST) to quantify cross-county distributional inconsistencies; designed a greedy agglomerative category-pooling optimizer that collapsed 38 property-categories into 13 semantically coherent groups, cutting the combined JSD objective by 79%.
- Built a 6-view geospatial dashboard (React, TypeScript, MapLibre-GL, D3) across 58 California counties that surfaces Empirical Bayes shrinkage weights, Jensen-Shannon divergence, Moran's I autocorrelation scores, and CatBoost C2ST accuracy as synchronized choropleth and edge maps.

Probabilistic Hurricane Track Forecasting

Interactive Dashboard

- Implemented a constant-velocity Kalman filter (linear state-space model) for hurricane track forecasting using sequential Bayesian inference on 721,960 observations from 13,450 tropical cyclones spanning 183 years; learned process and observation noise covariances via MLE from an 80/20 storm-level train/test split, achieving open-loop forecast errors of 16 km RMSE at 6-hour and 84 km RMSE at 24-hour lead times.
- Evaluated generalization using a sliding-origin protocol across 2,690 held-out storms and found that a velocity-persistence null model outperformed the Kalman filter by 3–22% across all lead times; diagnosed the gap to Gaussian assumption violations heavy-tailed forecast errors.